Artificial Intelligence and Jobs: Evidence from US Commuting Zones

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Any opinions and conclusions expressed in this discussion do not necessarily represent the views of the Banque de France.

This paper: What it does

- Builds a simple model demonstrating that AI can automate jobs or complement workers
- Studies the effect of AI on employment at the local level (722 CZs) in the United States between 2000 and 2020
- Measures AI adoption leveraging the use of specialized softwares
  - Computer scientists expertise: identify a list of AI-related 54 softwares
  - O*NET database: identify 82 occupations for which each software is *in demand*, refined to 19 AI-related occupations
  - Proxy for local AI adoption:

$$AIado_{ct} = \frac{L^{AI}_{cT_1} - L^{AI}_{cT_0}}{L_{c2000}}$$

  - Instrumented by a Bartik-style *AI exposure*, given the industry composition in 1980:

$$AIexp_{cdt} = \sum_{i=1}^{I} \lambda_{cdi1980} \times \left(\frac{L^{AI}_{iT_1} - L^{AI}_{iT_0}}{L_{i2000}} \right)$$

  - Intuition: Technological progress leads to ↘ AI cost / ↗ AI productivity, AI adoption ↗ differentially across industries
    - ⋆ Initial CZ labor composition will define a local exposure to AI

This paper: What it finds

- Negative relationship (OLS & IV) between AI adoption and local employment

Table 3: Baseline Results

	(1) OLS	(2) First Stage	(3) Reduced Form	(4) Second Stage	(5) OLS	(6) First Stage	(7) Reduced Form	(8) Second Stage
$AIado_{cdt}$	-0.253***			-0.671***	-0.238**			-1.594***
	[0.080]			[0.157]	[0.104]			[0.377]
$AIexp_{cdt}$		9.695***	-6.507***			9.599***	-15.298***	
		[1.128]	[1.447]			[1.813]	[3.279]	
Census Division FE	yes	yes	yes	yes	yes	yes	yes	yes
Decade FE	yes	yes	yes	yes	yes	yes	yes	yes
Control Variables	no	no	no	no	yes	yes	yes	yes
Kleibergen–Paap F-stat.	-	-	-	73.8	-	-	-	28.0
Obs.	1444	1444	1444	1444	1444	1444	1444	1444
R2	0.33	0.58	0.33	-	0.42	0.63	0.45	-

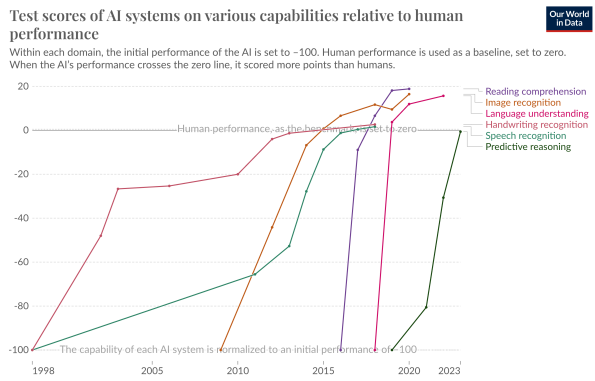
The sample consists of 722 CZs observed over two decades, 2000-2010 and 2010-2020. Each observation is a CZ x decade pair. The

- Heterogeneous effects
  - More (-) for middle-age workers than young workers
  - More (-) for low-skill workers
  - Monotonic relationship regarding wage: (-) for low-wage employees and (+) for high-wage employees

General Comment

- Very interesting paper on a hot topic
- Explores a wide range of outcomes to document heterogenity
- At a time of many policy discussions around AI (UK safety summit '23 ; French AI Summit '25), this papers provids new estimates that fuel the debate regarding the effects of AI on employment

Comment #1: Measuring AI-related softwares and occupations



- AI was not performing so well in numerous tasks until 2015
- Pre-2015 was still AI basic development
  - Deepmind created in 2010
  - Amazon Alexa launched in 2014

Comment #1: Measuring AI-related softwares and occupations

Table C3: Software used to Identify the AI-Related Occupations

Amazon Redshift	GitHub	Oracle PL/SQL
Amazon Simple Storage Service S3	Go	PHP
Amazon Simple Storage Service S4	JavaScript	Perl
Amazon Web Services AWS CloudFormation	JavaScript Object Notation JSON	PostgreSQL
Amazon Web Services AWS software	Jenkins CI	Python
Ansible Software	Kubernetes	Ruby
Apache Hadoop	Microsoft .NET Framework	Scala
Apache Hive	Microsoft Azure software	Selenium
Apache Kafka	Microsoft PowerShell	ServiceNow
Apache Spark	Microsoft SQL Server	Splunk Enterprise
Atlassian Confluence	Microsoft SQL Server Reporting Services SSRS	Spring Boot
Atlassian JIRA	MongoDB	Spring Framework
Bash	NoSQL	Structured query language SQL
C	Node.js	Transact-SQL
C#	Objective C	TypeScript
C++	Oracle Database	UNIX
Docker	Oracle Java	Vue.js
Git	Oracle Java 2 Platform Enterprise Edition J2EE	jQuery

- Software requirement based on **today** computer scientists **expertise**
- First limit: Softwares for AI activities might have been other softwares during the 2000s
  - In other words, was a AI developper using Python in 2005?

Comment #1: Measuring AI-related softwares and occupations

Table 1: AI-Related Occupations

Computer and Information Research Scientists	Mathematicians
Computer Network Architects	Network and Computer Systems Administrators
Computer Network Support Specialists	Operations Research Analysts
Computer Occupations, All Other	Software Developers
Computer Programmers	Software Quality Assurance Analysts and Testers
Computer Systems Analysts	Statistical Assistants
Computer User Support Specialists	Statisticians
Data Scientists	Web and Digital Interface Designers
Database Administrators	Web Developers
Database Architects	

Occupations are classified according to the 6-digit level of the 2018 Standard Occupational Classification.

- Second limit: Softwares for AI activities might have been useful for other ICT technologies adoption during the 2000s
  - Was a Web Developper using PHP and jQuery in 2010 adopting AI or building a website?
- Measure of AI adoption seems more ICT-related than AI-related, especially in the past
- To adress this second limit, the authors control for ICT equipment intensities (software, computers)
  - But does it really relate to AI vs. ICT? Significant penetration of Microsoft Office for cleck workers at that time
- Robustness focusing on Data Scientists seems insufficient because the same limit arises
- Need a defintion of a narrower measure of AI activities

Comment #2: AI adoption or AI development?

- The paper aims at studying the effect of AI *adoption* on employment through the lense of AI-related softwares
- But it seems to target AI *development* rather than large scale AI adoption
- For example, the adoption of AI in manufacturing is not only done through software-related occupation: an AI-assisted robot will be used by industrial workers (capital-embodied technical change)

Comment #3: Literature and micro channels

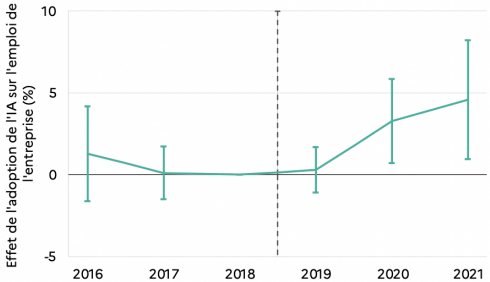
- Babina et al. (2024)
  - Build a measure of firms' investment in AI based on the intensity of their hiring in AI skills using resume and job offers
  - Find a positive effect on employment at (i) firm level and (ii) industry level

	Log Sales		Log Employment	
	(1)	(2)	(3)	(4)
Δ Share AI Workers	0.190***	0.220***	0.205***	0.260***
	(0.067)	(0.062)	(0.071)	(0.069)
NAICS2 FE	Y	Y	Y	Y
Controls	N	Y	N	Y
Observations	275	275	275	275

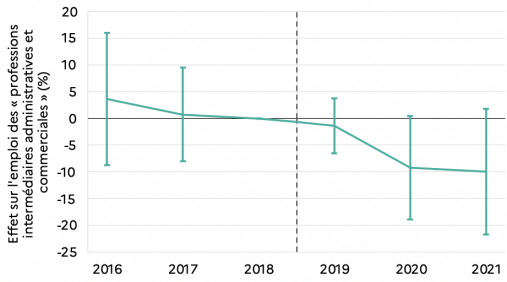
- Acemoglu et al. (2022)
  - No discernible relationship between AI exposure and employment at the industry level
- Empirical approaches close to this paper but opposite/different results
- How to reconcile?

Comment #3: Literature and micro channels

- Understanding the micro channels behind might be useful
  - Study outcomes on firm performance & firms dynamics: sales, entry, exit, etc.
  - Document the effect on firm and aggregate it at the local labor market level [Another paper?]



(a) Total employment



(b) Administrative and marketing occupations

Comment #4: IV estimates

- Bartik-style instrument of *AI exposure*, given the industry composition in 1980:

$$AIexp_{cdt} = \sum_{i=1}^{I} \lambda_{cdi1980} \times \left(\frac{L_{it_1}^{AI} - L_{it_0}^{AI}}{L_{i2000}} \right)$$

- *"The identifying assumption behind our approach is that the industry-level shifts are exogenous to shocks occurring in individual CZs."*
- The industry-level shifts of the ICT industry do not seem exogenous to shocks occuring in the Harvard/MIT CZ or the Silicon Valley CZ
- Table 4 deals with this concern, but would be useful to combine (i) excluding CZs with the highest $AIado_{cdt}$ and (ii) leave-one-out to build $AIexp_{cdt}$
- Or even use criterion (i) in the main regression